

Image Printing on Stones, Wood, and More

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Figure 1: Sample image printings on different objects.

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1 INTRODUCTION

Image engraving on caves is one of the oldest forms of expression and communication known to human beings, preceding structured written languages, and arguably even spoken ones. It is a surprise, then, and a shame, too, that in spite of all contemporary advances in computer graphics (CG), and our ability to simulate almost all visible entities, we are still unable to provide an easy solution for “printing” on stones and similar media used by our ancestors thousands of years ago. Such a gap in computer graphics of falling short of emulating artistic rather than photo-realistic expression was noted earlier by Salesin and colleagues [1994], who introduced Non-Photo-realistic Animation and Rendering (NPAR) as a new line of research in computer graphics, and launched an annual conference with that name. A lot of research followed in this new direction, but the overall scenario coincided with the main stream, photo-realistic rendering, in trying to bring the external world, art this time, to the computer display, rather than exporting computer-generated graphics to the external world. The common denominator of almost all fields of computer graphics today is the digital display, whether a computer monitor or a mobile screen or a VR goggle. In summary, computer graphics is forcing us into a virtual world rather than exporting contents to our physical world. One notable drawback of this is psychological, where a digitally-displayed image, for example, arguably feels *volatile*, far less precious than old-days hard-printed long-lasting photos.

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2 HALFTONING VS STIPLING

A primary reason for the mentioned shortcoming that CG is mostly locked inside a virtual world is that it is mostly built around a rectangular-grid representation, a 2D array of pixels, that is not well-suited to some kind of media. Indeed, a pixel array like that used in electronic displays might be easier to manufacture and more practical for dynamic updating, but is not necessarily the best for displaying a still image, especially a monotone one. Consider, for example, laser engraving on wood. Compared to an electronic display, this configuration has a limited tone (depth) resolution thanks to the limited control over a combustion process. Unlike the electronic display, however, the spacial resolution is not quite constrained. It is therefore more reasonable to consider *stippling*: arbitrarily placed dots, than halftoning, where the dots are placed over a finite-resolution grid. Computer generated stippling was introduced by Deussen et al. [2000] as a category of non-photo-realistic rendering. The lack of an efficient algorithm, however, restricted the application of the concept to stylization of images in computer displays, and we are not aware of an application as an alternative printing technology. Efficiency here refers to *bandwidth* rather than time; that is, the amount of information conveyed over a given number of dots, and the signal-to-noise ratio thereat.

3 GAUSSIAN BLUE NOISE

Quite recently, we [2022] introduced an advanced Gaussian-kernel-based Blue-Noise (GBN) algorithm for stippling and blue-noise sampling that is orders of magnitudes less noisier than the state of the art, opening the way for re-introducing stippling as an alternative practical encoding approach of images. Indeed, our stippling results are quite comparable to 256 grayscale images of the same information bits budget, as illustrated in Fig. 2. The essence of the Gaussian kernel is that it is separable yet isotropic, admitting Cartesian processing yet isotropic output, and the essence of improved bandwidth is that it admits a recognizable image rendering using only a small number of points and/or on a noisy medium.

4 DIRECT APPLICATION

Our GBN algorithm [2022] and a laser engraver are the main elements of our proposed method. We developed an improved kernel

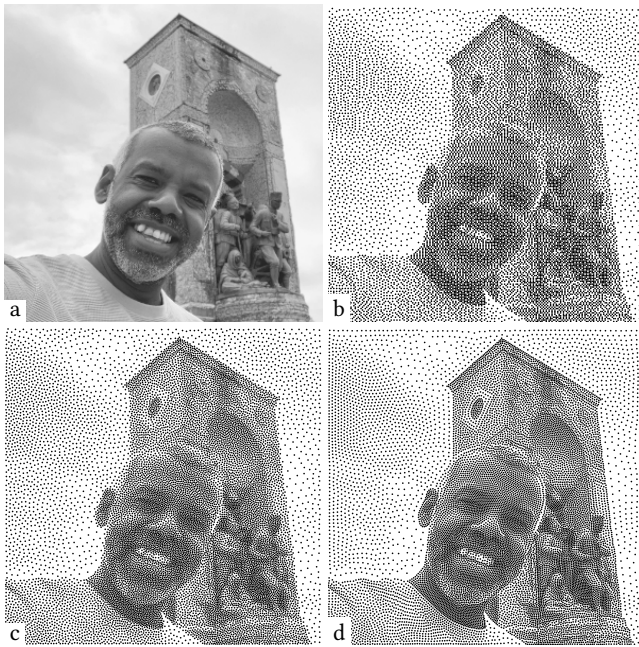


Figure 2: Transcoding an image representation, comparing (a) a gray scale image using $300 \times 300 \text{ pixel} \times 8\text{-bit tone} = 720\text{K}$ information bits, to (b, c, d) a $15\text{K-point} \times (24, 24)\text{-bit floating-point resolution}$, the same information bit count; obtained by (b) Floyd-Steinberg halftoning, (c) BNOT: then state of the art, and (d) our GBN. Note the excessive noise in BNOT and the sharp feature lines in GBN.

estimation algorithm to GBN that enables better handling of the boundaries. We also introduced an optional tone mapping that adds more emphasis to the edges, offering an improved artistic touch. With the stippling ready, we started our exploration and prototyping.

Some media, e.g., wood, leather, or fruits, readily render the stippling points when directly burnt by laser. For some other materials, such as metal, there are commercially available reagents that are applied to render the laser burns into a visible dark color. Such materials are relatively expensive, though, and are not quite suited for stones and tiles, which are our main target. We were surprised that some kinds of stones turn black upon laser burning, and that was actually the original motivation for this work. Our earlier prototypes targeted these types of stones, but we were soon not satisfied with the relatively poor quality and the excessive constraint, and started to search for a better approach.

5 STENCILS AND MASKS

At a later stage of our exploration, we considered using a permanent marker to make the laser marking more uniform than the natural burn effect, and also to mark a wider range of stones. Our earlier technique included three steps: (1) grind the stone to a smooth surface, (2) engrave the stippling with laser, (3) apply the marker to the whole area, and (4) grind again to remove the marker from the surface, leaving it only inside the engraved holes. While evidently

tedious, we were satisfied with the results. However, the method failed when tried on some porous materials such as tagua nuts, prompting further exploration. We eventually arrived at our favorite techniques of using an in-place stencil. Given a smooth surface of a stone, tile, mug, etc., our method comprises four steps: (1) apply a mask to the surface, (2) laser-engrave the stippling, adjusting the laser power such that it burns the mask and slightly etches the underlying surface, (3) apply a marker on top of the mask, and (4) remove the mask. We tried masking tape and ultraviolet car-window film as stencils, and obtained excellent results for both. The technique is quite mature now, and we have not experienced any failure applying it to ceramic tiles, for example. The mug in Fig. 1, for example, so far stood months of daily use.

The use of stencils is not confined to in place stencils, indeed. We have also made successful experiments with paper stencils to apply stippled images to garments and cake, but the concept is not yet mature as tile engraving, and we are still searching for better materials to use for the stencils.

6 CONCLUSION AND OUTLOOK

In this work we reintroduced stippling as a feasible encoding for image printing on many physical media including stones, metals, tiles, leather, mugs, etc. We presented working prototypes that open the way for a lot of follow-up research, including choice of ink, masking materials, laser settings, as well as improving the underlying stippling algorithm. The most obvious application of the presented work is personalization, where the improved bandwidth of the GBN stippling algorithm enables printing recognizable images on small items such as pen drives, and high-quality images on phones, for example. Another obvious application is printing precious images, e.g., wedding, on marble and tiles. Indeed, engraving on stone gives a feeling of long lasting. We used a low-end DIY laser engraver in all our experiments to make sure that the technique is accessible to a wide range of users. The GPU-based computation of the stippling is possibly the highest cost, which suggests a cloud-based business model. For example, a giant service provider could independently transcode and store the images to subscribers, while the actual printout could be provided by a local third party. Another stakeholder that might endorse the idea are manufacturers of laser engraving machines. In all cases, we plan to provide our code for free, and hope to build a community around the technique to develop standards and best practices.

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